

Build advanced and highly reliable industrial Ethernet communication system



ISM5012D Managed Industrial Ethernet Switch

· Public Utility

· Smart Manufacturing

· Smart City

· Smart Energy

1. Product Overview

The ISM series of managed industrial Ethernet switches is specifically designed to meet the demands of harsh industrial environments such as power, transportation, and industrial control applications.

Product Features:

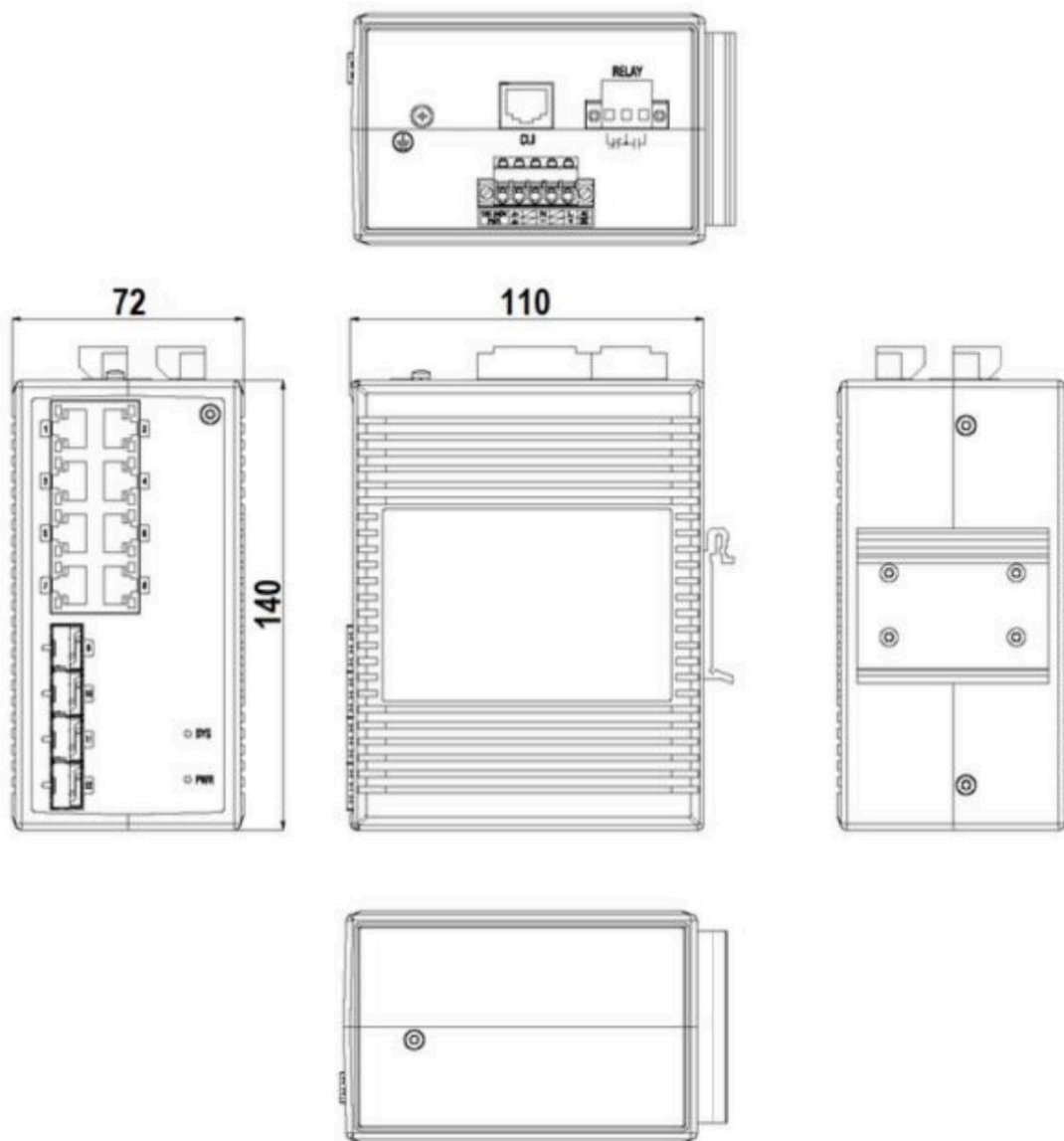
- **Reliability & Resilience:** Fanless design, IP40 protection rating, dust and dirt resistance, and wide temperature operation (-40 °C ~ 85 °C (-40 °F ~ 185 °F)).
- **Redundancy Protocols:** Supports STP/RSTP/MSTP ring redundancy protocols, offering flexible choices for intricate industrial networks.
- **Network Management:** Supports SNMP for integrated network management and RMON for effective network monitoring and fault prediction.
- **Enhanced Security & Control:** Port-based VLAN, IEEE 802.1Q VLAN, GVRP, IGMP Snooping, and QoS to process critical data.
- **Linear Industrial Switches:** High performance, flexible linear switches catering perfectly to industrial deployments.

Key Technical Specifications

Parameter	Specification
Type	Layer 2 Managed Industrial Ethernet Switch
Ports	4 x 100/1000BaseX SFP + 8 x 10/100/1000BaseT
Switching Performance	55 Gbps backplane bandwidth; 8K MAC table
Dimensions / Weight	72 mm x 140 mm x 110 mm; 1 kg (2.2 lb)
Power	18~60 VDC dual redundant input
Environment	-40 °C ~ +85 °C (-40 °F ~ +185 °F) operating; IP40

Parameter	Specification
EMC	FCC Part 15 Class A; IEC 61000-4-2/3/4/5/6/9
Management	Web, CLI, SNMPv1/v2c/v3, RMON; STP/RSTP/MSTP, IGMP Snooping, VLAN
Certifications	CE, FCC, IEC61850-3

2. Product Dimensions



ISM5012D

Note:

1. All dimensions are in millimeters (mm), with inches (in) in parentheses.

2. All dimensions are approximate and **for reference only**.
3. The dimensions shown in the figure **shall not be used for production or processing**.
4. Dimensions must comply with part and manufacturing tolerance requirements.
5. Dimensions are subject to change without notice.

3. Technical Specifications

3.1 Protocol Compliance List

Category/Parameter	Specification
IEEE Standards	
IEEE 802.3	CSMA/CD method and physical Layer specifications
IEEE 802.1p	Priority Queuing
IEEE 802.1q	VLAN tagging
IEEE 802.1d	Spanning Tree Algorithm
IEEE 802.1w	Rapid Spanning Tree
IEEE 802.1s	Multiple Spanning Tree
IEEE 802.3ac	VLAN Tagging
IEEE 802.1x	Authentication
IEEE 802.3ad	Link Aggregation
IEEE 802.3x	Flow Control
IEEE 802.3	Ethernet
IEEE 802.3u	Fast Ethernet
IEEE 802.3z	Gigabit Ethernet

Category/Parameter	Specification
IEEE 802	Networks
RFC Standards	
RFC 213	DHCP Server
RFC 768	UDP
RFC 791	IP
RFC 792	ICMP
RFC 793	TCP
RFC 826	ARP
RFC 854	Telnet Client & Server
RFC 862	Echo Protocol
RFC 863	Discard Protocol
RFC 868	Time Protocol
RFC 1027	Using ARP to Implement Transparent Subnet Gateways
RFC 1059, 1119	NTPv1/2
RFC 1112	IGMP
RFC 1191	Path MTU Discovery
RFC 1403	BGP OSPF Interaction
RFC 1542	Bootstrap Extensions & DHCP
RFC 1851	The ESP Triple DES Transform
RFC 1994	PPP Challenge Handshake Authentication Protocol (CHAP)
RFC 2068	HTTP

Category/Parameter	Specification
RFC 2138	RADIUS
RFC 2139	RADIUS Accounting
RFC 2236	IGMPv2
RFC 2474	DiffServ Precedence
RFC 2475	DiffServ Core and Edge Router Functions
RFC 2597	DiffServ Assured Forwarding
RFC 2598	DiffServ Expedited Forwarding
RFC 2865	Remote Authentication Dial In User Service (RADIUS)
RFC 3046	DHCP Relay Agent Information Option
RFC 3222	Forwarding Information Base (FIB)
Other Protocols	
GMRP	GARP
GVRP	GARP
SSH2	Secure Shell 2
IGMP	Snooping
SNMPv3	Supported

3.2 Hardware Specifications

Category/Parameter	Specification
Physical Performance	
Enclosure	Fully enclosed seamless metal enclosure

Category/Parameter	Specification
Dimensions (W × D × H)	72 mm × 140 mm × 110 mm
Weight	1 kg (2.2 lb)
Mounting Method	DIN-rail mounting
Cooling Method	Fanless cooling
Ingress Protection	IP40
Storage Temperature	-40 °C ~ +85 °C (-40 °F ~ +185 °F)
Operating Temperature	-40 °C ~ +85 °C (-40 °F ~ +185 °F)
Humidity	5 ~ 95% (non-condensing)
Hardware Performance	
Backplane Bandwidth	55 Gbps
Transmission Mode	Parallel Storage Forwarding
MAC Table Size	8K
Packet Buffer Size	4 Mbits
Exchange Rate	148,800 pps/100M ports; 1,488,000 pps/1000M ports
Power Parameters	
Input Voltage	18~60 VDC dual redundant input
Overload Current Protection	Supported
Reverse Polarity Protection	Supported

Category/Parameter	Specification
Electromagnetic Characteristics	
EMI	FCC 47 CFR Part 15 Class A; EN55022 Class A
EMS	IEC(EN)61000-4-2, Class 4 IEC(EN)61000-4-3, Class 3 IEC(EN)61000-4-4, Class 4 IEC(EN)61000-4-5, Class 4 IEC(EN)61000-4-6, Class 3 IEC(EN)61000-4-9, Class 5
Mechanical Characteristics	
Shock	IEC60068-2-27
Freefall	IEC60068-2-31
Vibration	IEC60068-2-6
Certifications	
Certifications	CE, FCC, IEC61850-3
Quality Assurance	
Warranty Period	5 years
MTBF	35 years

4. Software Specifications

Software Functions	Specification
Redundancy	STP, MSTP, RSTP, Port Trunking
Management Mode	Browser, Serial Port, STD-17 MIB-II, STD-58 SMlv2, STD-59 RMON, STD-62 SNMPv3, SNMPv2c, SNMPv1
Time Synchronization	NTP
Diagnostic Mode	Indicator light, Journal File, RMON, Port Mirroring, TRAP
Others	4K VLANs, IPv4/IPv6 multicast, storm control, support Jumbo Frame

5. Ordering Guide

Model	Description
ISM5012D-P-4GSFP-8GT-24	12-port Layer 2 managed Industrial Switch. 4 * 100/1000 BaseX SFP Ports and 8 * 10/100/1000BaseT Ports. Isolated Dual 18-60VDC Power Inputs. IP40 Protection Class.

6. Contact Us

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